

OS PYQ Traps & System Utilities

Lesson 52 — Resolving the Classic OS Traps

JKSSB Computer Awareness

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This Lesson Is a Trap Drill

- ▶ The computer section of JKSSB papers reuses a small set of **near-neighbour** terms.
- ▶ A wrong option is usually not random — it is the **closest confusion** to the right answer.
- ▶ In this lesson every question is built around one of those real trap patterns.
- ▶ **Read each option as a claim to be tested**, not merely as a phrase to recognise.
- ▶ Keep one fact per line and preserve the exact English term.

Trap drill

The wrong option is usually the “sounds right” neighbour — check the exact wording before you choose.

The Operating System: What It Is

- ▶ The **operating system (OS)** is **system software**.
- ▶ It provides a **user-friendly interface** between the user and the hardware (PYQ-21).
- ▶ It manages hardware resources: processor, memory, storage, and input/output devices.
- ▶ It provides services to application programs.
- ▶ Examples: Windows, Linux, UNIX, MS-DOS, Android, macOS.

Definition

The OS is the system software that controls the hardware and offers a user interface and services to programs.

Where the OS Lives While Running

- ▶ An OS is stored permanently on **secondary storage** (the disk).
- ▶ While the computer is **running**, the OS is loaded into **primary memory (RAM)** (PYQ-22).
- ▶ The CPU can execute OS code only from main memory.
- ▶ Power off clears RAM; the OS is reloaded from disk at the next boot.

Trap alert

The OS is not executed from secondary storage. At run time it resides in primary memory.

Three Software Categories

Category	Role	Examples
System software	Runs and manages the machine	OS, device drivers, utility programs, translators
Application software	Does a user's task	MS Word, Excel, a browser
Utility software	Maintains and optimises the system	Disk Cleanup, Disk Defragmenter, antivirus

Utility software is a maintenance category closely tied to system software.

Device Drivers Are Software

- ▶ A **device driver** is a small **software program** that lets the OS talk to a specific hardware device.
- ▶ It translates OS commands into device-specific instructions.
- ▶ Every device — printer, keyboard, graphics card — needs a matching driver.
- ▶ The driver is software; the device it controls is hardware.

Trap alert

A device driver is **software**, not hardware (TRAP-10). The OS itself is **system software**.

DMA: Transferring Data Without the CPU

- ▶ **DMA** stands for **Direct Memory Access**.
- ▶ A DMA controller transfers data directly between an **I/O device** and **main memory**.
- ▶ The CPU does **not** handle each individual byte of the transfer — it is bypassed.
- ▶ The CPU starts the transfer and is notified (by interrupt) when it finishes.
- ▶ This frees the CPU for other work and speeds up bulk transfers.

Definition

DMA is a method in which a controller moves data between a device and main memory without the CPU's continuous involvement.

DMA Trap Alert

- ▶ DMA moves data between **an I/O device and main memory**, not between the CPU and the device.
- ▶ The CPU is **bypassed** during the transfer, not made faster.
- ▶ DMA does **not** increase RAM size and does not encrypt data.
- ▶ The CPU still initiates and completes the transfer.

Trap alert

“DMA = the CPU does the copying faster” is wrong. DMA removes the CPU from the per-byte copying.

RTOS: Real-Time Operating System

- ▶ **RTOS** stands for **Real-Time Operating System**.
- ▶ It is designed so that tasks finish within **fixed time deadlines**.
- ▶ It is used in embedded and control systems: machinery, medical devices, vehicle electronics.
- ▶ Chosen for timing guarantees, not for word processing or graphics design.

Definition

An RTOS is an operating system that guarantees a response within a bounded time.

DOS, UNIX, and Linux Distributions

- ▶ **DOS = Disk Operating System**; a command-line OS (MS-DOS).
- ▶ **UNIX** is a multi-user, multitasking operating system.
- ▶ **Linux** is an open-source, UNIX-like OS; it comes in **distributions**.
- ▶ Common Linux distributions: **Ubuntu**, **Fedora**, **Debian**, **Linux Mint**.
- ▶ Windows and macOS are operating systems too, but they are not Linux distributions.

Exact OS Recognition

- ▶ Ask first: does the option actually name an **operating system**?
- ▶ Operating systems: **Windows**, **Linux**, **UNIX**, **MS-DOS**, **Android**, **macOS**.
- ▶ **Not** operating systems: MS Word, MS Excel, MS Access, Chrome, Firefox — these are applications.
- ▶ MS Access is a database application, not an operating system.

Trap alert

A familiar brand name alone does not make a system an OS — check whether it is system software.

Internal vs External Hardware

Location	Meaning	Examples
Internal	Inside the system unit / cabinet	motherboard, CPU, RAM, hard disk
External (peripheral)	Outside the system unit, joined by ports	monitor, keyboard, mouse, printer, scanner

The classification depends on position relative to the system unit.

The Windows Run Box and `winword`

- ▶ The Windows **Run** box (Windows + R) starts a program when you type its command.
- ▶ Typing `winword` opens **MS Word**.
- ▶ `notepad` opens Notepad; `wordpad` opens WordPad.
- ▶ `cmd` opens the Command Prompt.
- ▶ Match the exact command to the exact program.

Trap alert

`winword` = MS Word. Do not confuse it with `wordpad` (a different editor) or `notepad`.

Trap Alert: Terms to Keep Straight

- ▶ Device driver = **software**; OS = **system software** (TRAP-10).
- ▶ DMA moves data between a device and main memory, **bypassing the CPU**.
- ▶ RTOS = **Real-Time Operating System** (deadline-driven).
- ▶ Utility = maintenance software; application = the user's task.
- ▶ Linux distributions: Ubuntu, Fedora, Debian; **MS Access is not an OS**.
- ▶ winword opens MS Word.

Lesson 52: Recall

- ▶ The OS is system software; it provides a user interface, manages resources, and runs from primary memory.
- ▶ Software categories: system, application, and utility.
- ▶ A device driver is **software**; the OS is **system software**.
- ▶ DMA = Direct Memory Access; data moves between device and main memory with the CPU bypassed.
- ▶ RTOS = Real-Time Operating System.
- ▶ DOS = Disk Operating System; Linux distributions include Ubuntu, Fedora, and Debian.
- ▶ Internal vs external hardware; winword opens MS Word.